Lecture 11: Normal QQ Plots, Probability Distribution Tests

IEOR E4709: Statistic Analysis and Time Series
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1 Quantile to Quantile (QQ) Plot

A Q-Q plot ("Q" stands for quantile) is a *probability plot*, that is a graphical method for comparing two probability distributions, usually the sample distribution function and a population distribution function. The idea is to plot the quantiles of population and empirical distribution against each other. A point (x, y) on the plot corresponds to one of the quantiles of the second distribution (y-coordinate) plotted against the same quantile of the first distribution (x-coordinate).

Consider i.i.d. samples $X_1, X_2, \ldots, X_n$ coming from the distribution function F. For each sample of size n, $\{X_1, \ldots, X_n\}$, an empirical probability distribution function F_n is constructed by assigning mass $1/n$ to each observation X_i. Hence, F_n may be represented as $F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i \leq x}$, for $-\infty < x < \infty$. For each fixed value x, $F_n(x)$ is a random variable when it is viewed as a function of the sample of size n. In fact $F_n(x)$ is a random distribution function. The Glivenko-Cantelli Theorem ensures that the CDF converges uniformly almost surely to the true distribution function, i.e., $\mathbb{P}\left(\sup_x |F_n(x) - F(x)| \to 0\right\} = 1$.

Consider order statistics $X_{(1)} \leq X_{(2)} \leq \cdots, \leq X_{(n)}$. In general, we want to test the hypothesis H_0 that the samples are from the distribution F. Under H_0, $F(X_i)$'s are uniformly distributed. It is well known that the distribution of k-order statistics of the uniform distribution is a beta distribution $beta(k, n+1-k)$, where n is the number of random samples (see, for instance, the book by Gentle, James E. (2009), Computational Statistics, Springer, p. 63). Recall that the mean of a beta distribution $beta(k, n+1-k)$ is $\frac{k}{n+1}$. Define $Y_i = F(X_i)$ and observe that the order statistics $Y_{(1)}, \ldots, Y_{(n)}$ are the same as the order statistics $F(X_{(1)}), \ldots, F(X_{(n)})$ because $F(x)$ is an increasing function. Hence, it follows that $\mathbb{E}\left[F(X_{(k)})\right] = \frac{k}{n+1}$.

With a large sample size, $F(X_{(k)}) \approx \frac{k}{n+1}$, so that $X_{(k)} \approx F^{-1}(\frac{k}{n+1})$.[1] Hence, if H_0 is true, we expect to observe a line if we plot $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ versus $F^{-1}(\frac{1}{n+1}), F^{-1}(\frac{2}{n+1}), \ldots, F^{-1}(\frac{n}{n+1})$. This is a plot of theoretical (population) quantiles ($F^{-1}(\frac{k}{n+1})$'s) vs empirical quantiles ($X_{(k)}$'s), and is called quantile-to-quantile plot (QQ plot). If the plot is not linear, it may show nonlinearity at the upper or lower ends, which may be an indication of the presence of outliers. If the nonlinearity shows

[1] We know that $F(X_{(k)})$ follows a beta distribution $beta(k, n+1-k)$. Therefore, we have $\mathbb{E}\left[F(X_{(k)})\right] = \frac{k}{n+1}$ and $Var\left(F(X_{(k)})\right) = \frac{k(n+1-k)}{(n+1)^2(n+2)}$. Chebyshev's inequality states that if we have an integrable random variable X with finite expected value μ and finite non-zero variance σ^2, then for any real number $c > 0$, we have $\mathbb{P}(|X-\mu| \geq c\sigma) \leq \frac{1}{c^2}$. Hence, let $\epsilon = c\sigma$, we have $\mathbb{P}(|F(X_{(k)}) - \frac{k}{n+1}| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} = \frac{k(n+1-k)}{(n+1)^2(n+2)\epsilon^2}$. As $n \to \infty$, $\mathbb{P}(|F(X_{(k)}) - \frac{k}{n+1}| \geq \epsilon) \leq \frac{k(n+1-k)}{(n+1)^2(n+2)\epsilon^2} \to 0$. And we conclude that for each k, $F(X_{(k)})$ converges to $\frac{k}{n+1}$ in probability as $n \to \infty$.

up at other points as well, one could question the validity of the assumption that the probability distribution function of the random variables X_i's is F.

1.1 Normal Quantile to Quantile (QQ) Plot

The *normal* Q-Q plot is obtained by plotting $y = X_{(k)}$ versus $x = F^{-1}\left(\frac{k}{n+1}; \mu, \sigma\right)$, where X_i are distributed according to a Gaussian distribution with mean μ and variance σ^2, that is $F(x; \mu, \sigma^2) = \Phi(\frac{x-\mu}{\sigma})$. Here, $\Phi(x)$ is the cumulative distribution function of a Gaussian evaluated at x. Let $z_k = \Phi^{-1}\left(\frac{k}{n+1}\right)$, $k = 1, \ldots, n$ be the theoretical quantiles of a standard Gaussian distribution. Then $F^{-1}(\frac{k}{n+1}; \mu, \sigma) = \mu + \sigma z_k$. So, if we plot $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ versus $z_1, z_2, \ldots, z_n$, and the $X_i's$ are sample from a normal distribution, then we expect a linear line with intercept μ and slope σ.

2 Statistical Distribution Tests

We discuss statistical tests designed to test the hypothesis that observed samples come from a specific distribution, including the normal distribution.

2.1 The Kolmogorov-Smirnov (KS) Test

Suppose we have i.i.d. observations $X_1, X_2, \ldots, X_n$, which come from a distribution F. The KS test is used to test the null hypothesis H_0 that the samples come from F versus the alternative hypothesis H_1 that the samples do not come from F. Here, we have $F(x) := \mathbb{P}(X_i \leq x)$. We define the empirical cdf by $F_n(x) = \mathbb{P}_n(X \leq x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{X_i \leq x}$, where $X = (X_1, \ldots, X_n)$, that counts the proportion of sample points below x. For any fixed point x, the strong law of large numbers implies that $F_n(x) \to F(x)$, i.e., the empirical distribution converges to the true distribution F. Moreover, the Glivenko-Cantelli theorem strengthened this result by showing that this approximation holds uniformly, i.e. $\sup_x |F_n(x) - F(x)| \to 0$, i.e. the largest difference between F_n and F goes to zero in probability. The key observation in the KS test is that the distribution of the supremum $\sup_x |F_n(x) - F(x)|$ does not depend on F. Specifically, it can be shown that

$$\mathbb{P}\left(\sqrt{n}(\sup_x |F_n(x) - F(x)|) \leq t\right) \to H(t),$$

where $H(t) := 1 - 2\sum_{i=1}^{\infty}(-1)^{i-1}e^{-2i^2t^2}$ is the CDF of the *Kolmogorov-Smirnov distribution*. Consider the following statistics

$$D_n = \sqrt{n}(\sup_x |F_n(x) - F(x)|)$$

The null hypothesis is rejected if the value of the statistics is large. Let α be the significance level of the test. We have

$$\alpha = \mathbb{P}\left(D_n \geq c | H_0\right)$$

Since, under H_0, the cumulative probability distribution function of D_n is $H(t)$, and the quantiles of $H(t)$ can be tabulated, we can find the threshold $c = c_\alpha$ from the tables.

2.2 Normality tests

A distinguishing characteristic of the normal distribution, relative to other distributions, is its symmetry and lightness of the tails (i.e., extreme outcomes are very unlikely to occur for a normally distributed random variable).

Skewness measures the lack of symmetry in a distribution. A distribution is symmetric if it looks the same to the left and right of the center point. Formally, the skewness of X is defined as $\gamma_1(X) = \mathbb{E}\left[\frac{(X-\mu)^3}{\sigma^3}\right]$. Since any normal distribution is symmetric around its mean μ, its skewness is zero. The sample skewness is defined by $S := \frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^3/(s_X)^3$, where $\bar{X}$ is the sample mean, and $s_X = \left[\frac{1}{n}\sum_{j=1}^{n}(X_j - \bar{X})^2\right]^{1/2}$ is the sample standard deviation.

Kurtosis is a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution. That is, data sets with high kurtosis tend to have heavy tails. Data sets with low kurtosis tend to have light tails. The kurtosis is defined by $\gamma_2(X) = \frac{\mathbb{E}\left[(X-\mu)^4\right]}{\sigma^4} - 3$. Any normal distribution has zero kurtosis. The sample kurtosis is $K := -3 + \frac{1}{n}\sum_{j=1}^{n}\frac{(X_j-\bar{X})^4}{(s_X^4)}$.

Jarque-Bera Test: Suppose $X_1, X_2, \ldots, X_n$ are normally distributed i.i.d. random variables, and n is fairly large. Then, we expect S and K to be close to zero. A statistic for testing normality is the *Jarque–Bera statistic* defined as

$$JB = \frac{n}{6}\left(S^2 + \frac{1}{4}K^2\right).$$

As n becomes large, if normality holds, then the distribution of JB converges to a χ^2-distribution with two degrees of freedom. Normality is rejected if JB is sufficiently large.

Shapiro-Wilk Test: The test statistic is

$$W := \frac{\left(\sum_{i=1}^{n} a_i X_{(i)}\right)^2}{\sum_{i=1}^{n}(X_i - \bar{X})^2},$$

where $\bar{X}$ is the sample mean and $X_{(i)}$, $i = 1, \ldots, n$, is the i-th order statistic. Each number a_i is computed from $\mathbb{E}\left[X_{(i)}\right]$. Normality is rejected if W is small.